

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING) Department of Computer Science and
Engineering**

Course Code: ITA0609

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Course Title: Machine Learning

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Unit I Analytical Assignments with SDGs

Assignment Question:

1. Explain predictive maintenance in industries using Machine Learning . (SDG 9 – Industry ,Innovation and Infrastructure)

1. Problem Statement

Industries often suffer unexpected machine failures that increase downtime, repair costs and production losses. Design a Machine Learning (ML) based predictive maintenance system that continuously monitors machine health using sensor data and predicts failures before they occur.

2. Input

- Temperature
- Vibration
- Pressure
- Motor current
- RPM
- Sound
- Operating hours
- Maintenance history
- Failure logs

3. Output

- Machine health status (Normal/Faulty)
- Failure probability
- Remaining Useful Life (RUL)
- Maintenance schedule
- Alerts to maintenance team

4. Introduction

Predictive Maintenance (PdM) is an Industry 4.0 maintenance strategy that combines IoT sensors, data analytics and Machine Learning. Instead of waiting for a machine to fail (reactive maintenance) or servicing it at fixed intervals (preventive maintenance), predictive maintenance analyzes real-time data and predicts future failures. This helps industries repair equipment only when necessary, improving efficiency and reducing costs.

5. Objectives

- Detect equipment faults early.
- Reduce unplanned downtime.
- Reduce maintenance and repair cost.
- Increase equipment life.
- Improve worker safety.
- Increase production efficiency.

6. System Architecture

Sensors → IoT Gateway → Database/Cloud → Data Preprocessing → Feature Extraction → ML Model Training → Failure Prediction → Maintenance Dashboard → Maintenance Engineer

7. Working Explanation

1. Sensor data is collected continuously from industrial equipment.
2. Noise and missing values are removed during preprocessing.
3. Important features such as average vibration and temperature trends are extracted.
4. Historical labelled data is used to train ML models.
5. The trained model predicts whether the machine is healthy or likely to fail.
6. If the failure probability exceeds a threshold, an alert is generated and maintenance is scheduled before breakdown.

8. Machine Learning Algorithms

Decision Trees are simple and easy to interpret. Random Forest improves accuracy by combining multiple trees. Support Vector Machines classify healthy and faulty conditions effectively. Neural Networks and LSTM models are suitable for complex patterns and time-series sensor data. XGBoost provides high prediction accuracy for industrial datasets.

9. Advantages

- Minimizes downtime
- Reduces maintenance cost
- Improves product quality
- Increases equipment lifespan
- Enhances worker safety
- Optimizes spare-part usage

- Improves productivity

10. Limitations

The system requires high-quality sensor data, initial installation cost is high, ML models require periodic retraining, and cybersecurity must be ensured for connected industrial systems.

11. Applications

Used in manufacturing plants, CNC machines, power plants, wind turbines, railway engines, aircraft engines, oil and gas industries, smart factories and healthcare equipment.

12. SDG 9

Predictive maintenance directly supports Sustainable Development Goal 9 (Industry, Innovation and Infrastructure). It promotes smart industries, reduces waste, improves infrastructure reliability, encourages innovation through AI and IoT, and supports sustainable industrial growth.

13. Real-Time Example

A factory motor normally operates at 60°C with low vibration. The ML model detects that vibration has been increasing for several days and predicts a bearing failure within one week. The maintenance team replaces the bearing during a planned shutdown, preventing an expensive production stoppage.

14. Conclusion

Predictive maintenance is one of the most important applications of Machine Learning in modern industries. By analyzing sensor data and predicting failures before they happen, it saves time, reduces cost, improves safety, increases productivity and supports the vision of Industry 4.0 and SDG 9.